TKU211202
Linear Algebra
Introduction (After Midterm Exam)

Husni Rois Ali, Dzuhri Radityo Utomo, Yohan Fajar Sidik

Department of Electrical and Information Engineering
Universitas Gadjah Mada

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Introduction

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Course : Linear Algebra
Course Credit : 3
Prerequisites : Vector and Matrix Theory
Instructors

- Dzuhri Radityo Utomo, , ST, M.E., Ph.D.
- Dr.-Ing Yohan Fajar Sidik, S.T., M.Eng.

Contact Person

- **Name**: Yohan Fajar Sidik
- **Contact**: yohanfajarsidik@ugm.ac.id (Email)
- It is highly recommended to conduct all communications (questions, etc.) via Email or eLOK.

Lecture

- **Time**: as in your Simaster.
- It will be conducted offline.

Lecture Materials and Textbooks

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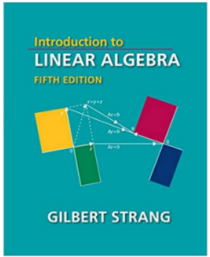
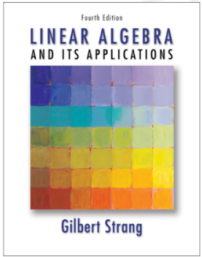
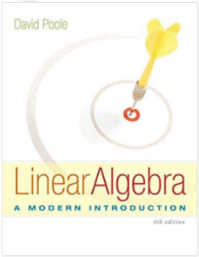
Lecture Materials

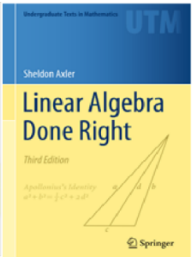
- LMS Platform: eLOK.
- All materials will be uploaded into Google Classroom/Elok.
- All assignments (Homeworks will be uploaded into Google Classroom/Elok, and all works should be submitted via eLOK)
- No submission via instructor's Email address of personal chat is allowed.

Lecture Materials and Textbooks









Textbooks

- Introduction to Linear Algebra by Gilbert Strang.
- Linear Algebra and Its Applications by Gilbert Strang.
- Linear Algebra by David Poole.
- Linear Algebra Done Right by Sheldon Axler.
- Introduction to Applied Linear Algebra Vectors, Matrices, and Least Squares by S. Boyd and L. Vandenberghe.

Course Workload

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Definition of 1 SKS

- Class Lecture: 50 minutes/weak.
- Individual Study: 60 minutes/weak.
- Homework: 60 minutes/week.

Definition of 3 SKS

- Class Lecture: 150 minutes/week.
- Individual Study: 180 minutes/week.
- Homework: 180 minutes/week.

Syllabus (After Midterm Exam)

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Singular Value Decomposition (SVD)

- SVD motivation
- SVD problem
- Geometry of SVD
- SVD for symmetric matrices

Linear Transformations

- The idea of linear transformation
- Linear transformations in matrix language
- Finding good basis

Complex Vectors and Matrices

- Complex number
- Hermitian and unitary

Learning Outcomes - After Midterm Exam

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LO2

Students are able to explain the concepts of eigenvalues, eigenvectors, and singular value decompositions (SVD), able to compute them.

LO3

Students are able to discuss the concept of linear transformation and translate this into matrix language.

LO4

Students are able to do mathematical calculations using complex vectors and matrices.

Learning Outcomes, Assessment, and Grading

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Assessment

1. Learning Outcomes (LO) will be assessed via Homework, Quiz, and Final Exam.
2. Quiz will be held during class **without any announcement** beforehand.

Grading (Tentative)

1. Before Midterm Exam: 50 %.
2. After Midterm Exam: 50 %.
   - Homework: 10-15 %.
   - Quiz: 10-15 %.
   - Final Exam: 25 %.

Softwares

1. Octave
2. Python

Academic Misconduct